Notebook

TRIG Degrees

Rundkorsel (Roundabout) - CAS-first readable solution

This notebook is structured as short alternating text and math cells.
Goal: explain every step clearly, not only compute answers.

Task (quoted, Danish)

> Projektopgaver: Rundkorsel
>
> Bestem cirklens ligning for vejbanens inderste cirkel,
> hvis koordinatsystemet anbringes i rundkorslens midte.
>
> Bestem cirklens ligning for den yderste cirkel,
> der gar igennem punkterne A (3,38) og B (26,25)
> i det indlagte koordinatsystem.
>
> Hvor lang tid vil det tage at kore en hel tur midt i rundkorslen
> med en anbefalet max hastighed pa 30 km/t?
>
> Bestem koordinaterne til punkterne C og D,
> nar det gives, at Kalundborgvej moder rundkorslen symmetrisk fra B til D.
> Vinklen imellem de to veje er 91 grader.
> Bestem herefter vinklen imellem de to tangenter til cirklen
> i punkterne A og C samt B og D.
>
> Bestem den korteste afstand imellem en bilist midt pa vejbanen
> i rundkorslen og den ventende bla bils midtpunkt P (43,30).
>
> Du skal nu projektere en ny vej,
> hvis linjeforing skal "ramme" rundkorslen fra nordostlig retning.
> Du skal bestemme denne linjeforings rette linje,
> bestemme punktet hvor linjeforingen rammer den yderste cirkel,
> samt de to anlobspunkter ved vejens side,
> der tangerer den yderste cirkel i rundkorslen.
> Du skal antage, at denne nye vej ankommer til rundkorslen
> som Kalundborgvej.

Assumptions used in this notebook:

- `rin = 15`, `rout = 25`, `rmid = 20` (meters).
- `91` is treated as the central angle `AOC`.
- Trig mode is `deg`, so `sin(91)` is in degrees.

1) Constants and circle equations in center coordinates

We start with constants and the three concentric circles in local center coordinates `(u,v)`.

Degrees equation

```
Ax := 3
```

$$Ax = 3$$

```
Ay := 38
```

$$Ay = 38$$

```
Bx := 26
```

$$Bx = 26$$

```
By := 25
```

$$By = 25$$

```
Px := 43
```

$$Px = 43$$

```
Py := 30
```

$$Py = 30$$

```
rin := 15
```

$$rin = 15$$

```
rout := 25
```

$$rout = 25$$

```
rmid := 20
```

$$rmid = 20$$

```
u^2 + v^2 = rin^2
```

$$u^2 + v^2 = 15^2$$

```
u^2 + v^2 = rout^2
```

$$u^2 + v^2 = 25^2$$

```
u^2 + v^2 = rmid^2
```

$$u^2 + v^2 = 20^2$$

2) Outer circle in embedded coordinates

From points `A`, `B`, and radius `rout`, there are two candidate centers.
We pick the one with larger `y` (the black dot in the figure is above A and B).

Degrees equation

```
dx := Bx - Ax
```

$$dx = 26 - 3$$

$$dx = 23$$

```
dy := By - Ay
```

$$dy = 25 - 38$$

$$dy = -13$$

```
Dab := (dx^2 + dy^2)^0.5
```

$$Dab = \left((-13)^2 + 23^2\right)^{0.5}$$

$$Dab = 26.4196896272458$$

```
mx := (Ax + Bx)/2
```

$$mx = \frac{3 + 26}{2}$$

$$mx = \frac{29}{2}$$

```
my := (Ay + By)/2
```

$$my = \frac{25 + 38}{2}$$

$$my = \frac{63}{2}$$

```
a := Dab/2
```

$$a = \frac{26.4196896272458}{2}$$

$$a = 13.2098448136229$$

```
halfChordToCenter := (rout^2 - a^2)^0.5
```

$$halfChordToCenter = \left(-13.2098448136229^2 + 25^2\right)^{0.5}$$

$$halfChordToCenter = 21.2249852767911$$

```
ux := -dy / Dab
```

$$ux = \frac{13}{26.4196896272458}$$

$$ux = 0.492057256667902$$

```
uy := dx / Dab
```

$$uy = \frac{23}{26.4196896272458}$$

$$uy = 0.870562838720134$$

```
h1 := mx + ux*halfChordToCenter
```

$$h_1 = 0.492057256667902 \cdot 21.2249852767911 + \frac{29}{2}$$

$$h_1 = 24.9439080281144$$

```
k1 := my + uy*halfChordToCenter
```

$$k_1 = 0.870562838720134 \cdot 21.2249852767911 + \frac{63}{2}$$

$$k_1 = 49.9776834343563$$

```
h2 := mx - ux*halfChordToCenter
```

$$h_2 = \frac{29}{2} - 0.492057256667902 \cdot 21.2249852767911$$

$$h_2 = 4.05609197188555$$

```
k2 := my - uy*halfChordToCenter
```

$$k_2 = \frac{63}{2} - 0.870562838720134 \cdot 21.2249852767911$$

$$k_2 = 13.0223165656437$$

```
h1N := N(h1, 8)
```

$$h1N = 24.943908$$

```
k1N := N(k1, 8)
```

$$k1N = 49.977683$$

```
h2N := N(h2, 8)
```

$$h2N = 4.056092$$

```
k2N := N(k2, 8)
```

$$k2N = 13.022317$$

```
(h1N, k1N)
```

$$(24.943908,\ 49.977683)$$

```
(h2N, k2N)
```

$$(4.056092,\ 13.022317)$$

```
h0 := h1N
```

$$h_0 = 24.943908$$

```
k0 := k1N
```

$$k_0 = 49.977683$$

```
(x - h0)^2 + (y - k0)^2 = rout^2
```

$$(x - 24.943908)^2 + (y - 49.977683)^2 = 25^2$$

3) Find C from angle 91 and D by symmetry

C is built by rotating vector OA by +91 degrees (east-side solution).
D is reflection of B across line OC.

Degrees expression

```
vAx := Ax - h0
```

$$vAx = 3 - 24.943908$$

$$vAx = -21.943908$$

```
vAy := Ay - k0
```

$$vAy = 38 - 49.977683$$

$$vAy = -11.977683$$

```
vCpx := vAx*cos(91) - vAy*sin(91)
```

$$vCpx = -21.943908\left(-\cos\left(\frac{89\pi}{180}\right)\right) - -11.977683\sin\left(\frac{89\pi}{180}\right)$$

$$vCpx = 21.943908\sin\left(\frac{\pi}{180}\right) + 11.977683\cos\left(\frac{\pi}{180}\right)$$

```
vCpy := vAx*sin(91) + vAy*cos(91)
```

$$vCpy = -21.943908\sin\left(\frac{89\pi}{180}\right) - 11.977683\left(-\cos\left(\frac{89\pi}{180}\right)\right)$$

$$vCpy = -21.943908\cos\left(\frac{\pi}{180}\right) + 11.977683\sin\left(\frac{\pi}{180}\right)$$

```
vCmx := vAx*cos(91) + vAy*sin(91)
```

$$vCmx = -11.977683\sin\left(\frac{89\pi}{180}\right) - 21.943908\left(-\cos\left(\frac{89\pi}{180}\right)\right)$$

$$vCmx = -11.977683\cos\left(\frac{\pi}{180}\right) + 21.943908\sin\left(\frac{\pi}{180}\right)$$

```
vCmy := -vAx*sin(91) + vAy*cos(91)
```

$$vCmy = -11.977683\left(-\cos\left(\frac{89\pi}{180}\right)\right) + 21.943908\sin\left(\frac{89\pi}{180}\right)$$

$$vCmy = 11.977683\sin\left(\frac{\pi}{180}\right) + 21.943908\cos\left(\frac{\pi}{180}\right)$$

```
Cplusx := h0 + vCpx
```

$$Cplusx = \left(21.943908\sin\left(\frac{\pi}{180}\right) + 11.977683\cos\left(\frac{\pi}{180}\right)\right) + 24.943908$$

$$Cplusx = 21.943908\sin\left(\frac{\pi}{180}\right) + 11.977683\cos\left(\frac{\pi}{180}\right) + 24.943908$$

```
Cplusy := k0 + vCpy
```

$$Cplusy = \left(-21.943908\cos\left(\frac{\pi}{180}\right) + 11.977683\sin\left(\frac{\pi}{180}\right)\right) + 49.977683$$

$$Cplusy = -21.943908\cos\left(\frac{\pi}{180}\right) + 11.977683\sin\left(\frac{\pi}{180}\right) + 49.977683$$

```
Cminusx := h0 + vCmx
```

$$Cminusx = \left(-11.977683\cos\left(\frac{\pi}{180}\right) + 21.943908\sin\left(\frac{\pi}{180}\right)\right) + 24.943908$$

$$Cminusx = -11.977683\cos\left(\frac{\pi}{180}\right) + 21.943908\sin\left(\frac{\pi}{180}\right) + 24.943908$$

```
Cminusy := k0 + vCmy
```

$$Cminusy = \left(11.977683\sin\left(\frac{\pi}{180}\right) + 21.943908\cos\left(\frac{\pi}{180}\right)\right) + 49.977683$$

$$Cminusy = 11.977683\sin\left(\frac{\pi}{180}\right) + 21.943908\cos\left(\frac{\pi}{180}\right) + 49.977683$$

```
N(Cplusx)
```

$$37.3027412033663$$

```
N(Cplusy)
```

$$28.2461569519854$$

```
N(Cminusx)
```

$$13.3510228718687$$

```
N(Cminusy)
```

$$72.1272886965326$$

```
Cx := Cplusx
```

$$Cx = 21.943908 \sin\left(\frac{\pi}{180}\right) + 11.977683 \cos\left(\frac{\pi}{180}\right) + 24.943908$$

```
Cy := Cplusy
```

$$Cy = -21.943908 \cos\left(\frac{\pi}{180}\right) + 11.977683 \sin\left(\frac{\pi}{180}\right) + 49.977683$$

```
vBx := Bx - h0
```

$$vBx = 26 - 24.943908$$

$$vBx = 1.056092$$

```
vBy := By - k0
```

$$vBy = 25 - 49.977683$$

$$vBy = -24.977683$$

```
dxC := Cx - h0
```

$$dxC = (-1)\, 24.943908 + \left(21.943908 \sin\left(\frac{\pi}{180}\right) + 11.977683 \cos\left(\frac{\pi}{180}\right) + 24.94390\right.$$

$$dxC = 21.943908 \sin\left(\frac{\pi}{180}\right) + 11.977683 \cos\left(\frac{\pi}{180}\right)$$

```
dyC := Cy - k0
```

$$dyC = (-1)\,49.977683 + \left(-21.943908\cos\left(\frac{\pi}{180}\right) + 11.977683\sin\left(\frac{\pi}{180}\right) + 49.9776\right.$$

$$dyC = -21.943908\cos\left(\frac{\pi}{180}\right) + 11.977683\sin\left(\frac{\pi}{180}\right)$$

```
dotBC := vBx*dxC + vBy*dyC
```

$$dotBC = 1.056092\left(21.943908\sin\left(\frac{\pi}{180}\right) + 11.977683\cos\left(\frac{\pi}{180}\right)\right) - 24.977683\left(-\right.$$

$$dotBC = -276.0\sin\left(\frac{\pi}{180}\right) + 560.75752\cos\left(\frac{\pi}{180}\right)$$

```
dotCC := dxC^2 + dyC^2
```

$$dotCC = \left(21.943908\sin\left(\frac{\pi}{180}\right) + 11.977683\cos\left(\frac{\pi}{180}\right)\right)^2 + \left(-21.943908\cos\left(\frac{\pi}{180}\right.\right.$$

$$dotCC = 625.0$$

```
projx := (dotBC/dotCC)*dxC
```

$$projx = \frac{\left(-276.0\sin\left(\frac{\pi}{180}\right) + 560.75752\cos\left(\frac{\pi}{180}\right)\right)\left(21.943908\sin\left(\frac{\pi}{180}\right) + 11.977683\right.}{625.0}$$

$$projx = 7.1994967\sin\left(\frac{\pi}{90}\right) + 0.52804597 + 10.218476\cos\left(\frac{\pi}{90}\right)$$

```
projy := (dotBC/dotCC)*dyC
```

$$projy = \frac{\left(-276.0\sin\left(\frac{\pi}{180}\right) + 560.75752\cos\left(\frac{\pi}{180}\right)\right)\left(-21.943908\cos\left(\frac{\pi}{180}\right) + 11.97768\right.}{625.0}$$

$$projy = -12.488842 - 7.1994967\cos\left(\frac{\pi}{90}\right) + 10.218476\sin\left(\frac{\pi}{90}\right)$$

```
vDx := 2*projx - vBx
```

$$vDx = (-1)\,1.056092 + 2\left(7.1994967\sin\left(\frac{\pi}{90}\right) + 0.52804597 + 10.218476\cos\left(\frac{\pi}{90}\right)\right.$$

$$vDx = -2.9802322\cdot 10^{-8} + 14.398993\sin\left(\frac{\pi}{90}\right) + 20.436952\cos\left(\frac{\pi}{90}\right)$$

```
vDy := 2*proj y - vBy
```

$$vDy = 2\left(-12.488842 - 7.1994967\cos\left(\frac{\pi}{90}\right) + 10.218476\sin\left(\frac{\pi}{90}\right)\right) - -24.977683$$

$$vDy = -14.398993\cos\left(\frac{\pi}{90}\right) - 2.9802322\cdot 10^{-8} + 20.436951\sin\left(\frac{\pi}{90}\right)$$

```
Dx := h0 + vDx
```

$$Dx = \left(-2.9802322\cdot 10^{-8} + 14.398993\sin\left(\frac{\pi}{90}\right) + 20.436952\cos\left(\frac{\pi}{90}\right)\right) + 24.9439($$

$$Dx = 14.398993\sin\left(\frac{\pi}{90}\right) + 20.436952\cos\left(\frac{\pi}{90}\right) + 24.943908$$

```
Dy := k0 + vDy
```

$$Dy = \left(-14.398993\cos\left(\frac{\pi}{90}\right) - 2.9802322\cdot 10^{-8} + 20.436951\sin\left(\frac{\pi}{90}\right)\right) + 49.97768$$

$$Dy = -14.398993\cos\left(\frac{\pi}{90}\right) + 20.436951\sin\left(\frac{\pi}{90}\right) + 49.977683$$

```
N(Cx)
```

$$37.3027412033663$$

```
N(Cy)
```

$$28.2461569519854$$

```
N(Dx)
```

$$45.8709275101925$$

```
N(Dy)
```

$$36.3007007614685$$

```
checkC := simplify((Cx - h0)^2 + (Cy - k0)^2 - rout^2)
```

$$checkC = 0$$

```
checkD := simplify((Dx - h0)^2 + (Dy - k0)^2 - rout^2)
```

$$checkD = -1.2181372 \cdot 10^{-6} \cos\left(\frac{\pi}{90}\right) - 5.8207658 \cdot 10^{-7} - 8.5824681 \cdot 10^{-7} \sin\left(\frac{}{9}\right.$$

```
checkC
```

$$0$$

```
checkD
```

$$-1.2181372 \cdot 10^{-6} \cos\left(\frac{\pi}{90}\right) - 5.8207658 \cdot 10^{-7} - 8.5824681 \cdot 10^{-7} \sin\left(\frac{\pi}{90}\right) + 5.82($$

4) Angles between tangents

For points `A` and `C`: tangent angle is `180 - 91 = 89`.
For points `B` and `D`: compute central angle `BOD`, then tangent angle is `180 - BOD`.

Degrees expression

```
phiAC := 180 - 91
```

$$phiAC = 180 - 91$$

$$phiAC = 89$$

```
thetaBD := (180/pi) * acos((vBx*vDx + vBy*vDy)/rout^2)
```

$$thetaBD = \frac{180\,\mathrm{acos}\left(\frac{1.056092\left(-2.9802322\cdot 10^{-8}+14.398993\sin\left(\frac{\pi}{90}\right)+20.436952\cos\left(\frac{\pi}{90}\right)\right)-24.977683\left(-\right.}{25^2}\right.}{\pi}$$

$$thetaBD = \frac{180\,\mathrm{acos}\left(-0.79241767\sin\left(\frac{\pi}{90}\right)+1.1406704\cdot 10^{-9}+0.60997888\cos\left(\frac{\pi}{90}\right.\right.}{\pi}$$

```
phiBD := 180 - thetaBD
```

$$phiBD = 180 - \frac{180\,\mathrm{acos}\left(-0.79241767\sin\left(\frac{\pi}{90}\right)+1.1406704\cdot 10^{-9}+0.60997888\,\mathrm{cc}\right.}{\pi}$$

$$phiBD = -\frac{180\,\mathrm{acos}\left(-0.79241767\sin\left(\frac{\pi}{90}\right)+1.1406704\cdot 10^{-9}+0.60997888\cos\left(\frac{\pi}{90}\right.\right.}{\pi}$$

```
phiAC
```

$$89$$

```
N(thetaBD)
```

$$54.4120239288058$$

```
N(phiBD)
```

$$125.587976071194$$

5) Travel time and shortest distance to P

Lap length on midline is `2*pi*rmid`.
Speed is `30 km/h = 30000/3600 m/s`.
Shortest distance from point `P` to midline circle is `|OP - rmid|`.

Degrees expression

```
L := 2*pi*rmid
```

$$L = 2\pi 20$$

$$L = 40\pi$$

```
speed := 30000/3600
```

$$speed = \frac{30000}{3600}$$

$$speed = \frac{25}{3}$$

```
T := L / speed
```

$$T = \frac{40\pi}{\frac{25}{3}}$$

$$T = \frac{24\pi}{5}$$

```
OP := ((Px - h0)^2 + (Py - k0)^2)^0.5
```

$$OP = \left((43 - 24.943908)^2 + (30 - 49.977683)^2\right)^{0.5}$$

$$OP = 26.9282433875435$$

```
dmin := abs(OP - rmid)
```

$$dmin = 6.92824338754353$$

```
N(T)
```

$$15.079644737231$$

```
N(dmin)
```

$$6.92824338754353$$

6) New road from northeast (45 degrees)

Road centerline goes through `O=(h0,k0)` with slope `tan(45)=1`.
We reuse Kalundborg approach half-angle `alpha = angle(BOC)` to get edge contact points `E1`, `E2`.
Then we compute tangent lines at `E1`, `E2` in slope-intercept form (numeric).

Degrees *running...*

Click to edit.

7) Compact final summary

This final cell prints key numeric results in short lines.
We intentionally skip heavy plotting here to keep notebook layout compact on laptop screens.

Degrees *running...*

Click to edit.